

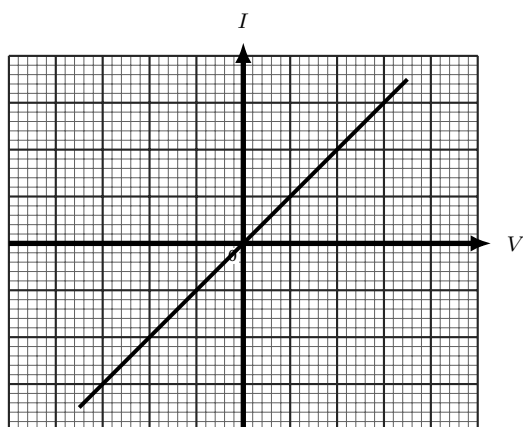
Name: _____
 Class: _____
 Date: _____

Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

1 · Ohmic conductors resistance stays the same



Graph A

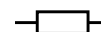
The shape: a straight line through the origin.

Double the pd, and the current doubles too. So $R = \frac{V}{I}$ comes out **the same everywhere on the line**.

$$I \propto V$$

Examples

- a **fixed resistor**
- a **metal wire** — as long as it stays cool



The condition: at a constant temperature. Let it get hot and it stops being ohmic.

Worked example — reading graph A twice

At $V = 1.0 \text{ V}$: read up, $I = 0.20 \text{ A}$. $R = \frac{V}{I} = \frac{1.0}{0.20} = 5.0 \Omega$

At $V = 3.0 \text{ V}$: read up, $I = 0.60 \text{ A}$. $R = \frac{V}{I} = \frac{3.0}{0.60} = 5.0 \Omega$

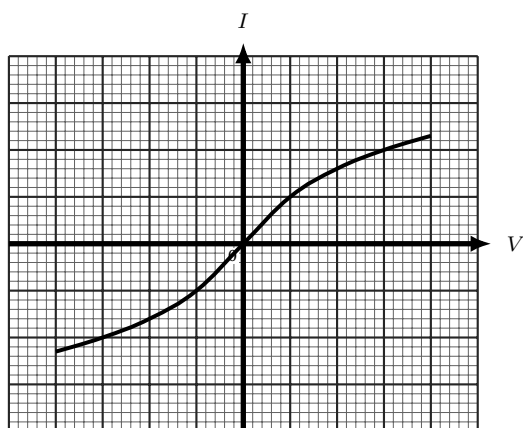
The same answer both times. That is what makes graph A **ohmic** — the resistance is 5.0Ω wherever you measure it.

Finding resistance from any graph

Pick a point on the line. Read V across. Read I up. Divide. $R = \frac{V}{I}$ — this works on every graph on this sheet.

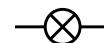
The check: do it again at a second point. Same answer → the resistance is **constant** (ohmic). Different answer → the resistance has **changed**.

The resistance of a **lamp**, **diode**, **thermistor** or **LDR** is **not** constant — so their graphs are not straight lines.



Graph B

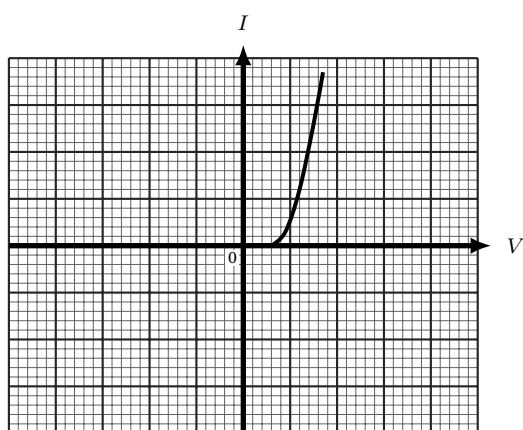
2a The filament lamp



The shape: bends over towards the V axis — the same way on *both* sides.

Why: bigger current $\rightarrow$ filament **hotter** $\rightarrow$ ions **vibrate more** $\rightarrow$ more **collisions** $\rightarrow$ **resistance increases**.

Examples: a torch bulb, a car headlamp, an old-style light bulb.



Graph C

2b The diode



The shape: flat along the axis one way, then rising steeply the other way.

Why: a diode lets current through in **one direction only**. Forwards, it conducts once the pd passes about 0.6 V. Backwards, its resistance is **very high**, so the current is almost zero.

Examples: the charging circuit in a phone, a solar panel, anything that must not be connected backwards.

Two more, for the exam

Thermistor — resistance **decreases** as it gets **hotter**. Used in a thermostat. · **LDR** — resistance **decreases** as it gets **brighter**. Used to switch street lights on at dusk.

3 · Questions graphs A, B and C are on the front

1. State what is meant by an **ohmic conductor**. [2]

A conductor whose resistance stays constant (at constant temperature), so the current through it is directly proportional to the pd across it.

2. Name the component that produced each graph. [3]

A: *fixed resistor*

B: *filament lamp*

C: *diode*

3. Use **graph B** to find the resistance of the lamp at $V = 1.0\text{ V}$ and at $V = 3.0\text{ V}$. Show your working, and say what has happened to the resistance. [4]

At $V = 1.0\text{ V}$, $I = 0.20\text{ A}$: $R = V/I = 1.0/0.20 = 5.0\ \Omega$.

At $V = 3.0\text{ V}$, $I = 0.40\text{ A}$: $R = V/I = 3.0/0.40 = 7.5\ \Omega$.

*The resistance has **increased**, because the filament is hotter at the higher pd. Accept readings within one small square.*

4. Jamie found the resistance for **graph A** by measuring its gradient, and wants to use the same method on **graph B**. State what Jamie should do instead for graph B, and explain why the method that worked on graph A does not carry across. [3]

*He should read **one point** off the curve and calculate $R = V/I$ at that point. Graph A is a straight line through the origin, so its gradient equals $1/R$ and happens to give the right answer there — but graph B is a curve, so it has no single gradient, and resistance is V/I rather than $\Delta I/\Delta V$ in any case. $R = V/I$ at a point works for both.*

5. Explain, in terms of the filament, **why** graph B bends. [3]

As the current increases the filament gets hotter. The metal ions vibrate more, so the electrons collide with them more often, and the resistance increases — so each extra volt gives less extra current.

6. Look at the **left-hand half** of the graphs, where the pd has been reversed. [4]

(a) Describe what **graph C** does, and explain why.

Almost no current flows — the line lies flat along the V axis. A diode conducts in one direction only; in reverse its resistance is very high.

(b) **Graph B** is symmetrical about the origin. What does that tell you about a filament lamp?

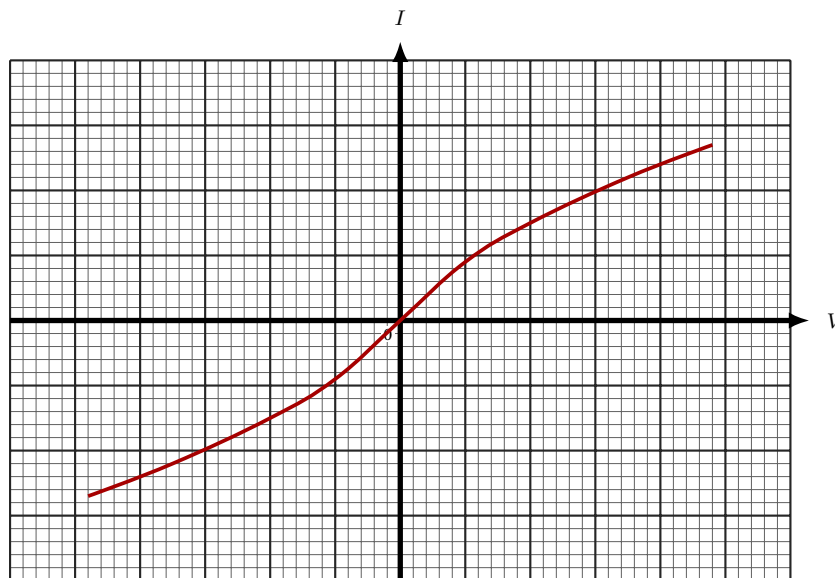
It behaves the same way whichever direction the current flows. It is non-ohmic, but it is not one-way.

7. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its graph, and explain in one sentence. [2]

it begins to bend over it stays perfectly straight it flattens along the axis

It is no longer at a constant temperature, so its resistance is no longer constant — the line starts to curve and it stops behaving ohmically.

8. Sketch the graph you would expect for a **filament lamp**. Use both halves of the axes. [3]



Through the origin; curving over towards the V axis; drawn in both halves. One mark each.